Sahaj Bhadja

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Summary

Software Engineer with 2+ years implementing **cloud-integrated infrastructure workflows**, **automation pipelines**, and **secure operational controls** using Python, AWS/Azure services, SQL, Docker, and CI/CD practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Implemented **cloud-backed data services** for 500+ users over **Azure SQL/PostgreSQL**, reducing latency from **1.2s to 350ms** through resource optimization and resilient service tuning.
- Built **validation and governance-style checks** for 5K+ monthly jobs, improving drift detection, operational reliability, and remediation speed for production workflow environments.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **automation-ready backend/frontend features** handling 10K+ daily requests, integrating APIs and secure ingestion flows to support scalable operational workloads.
- Improved **CI/CD delivery controls** and concurrent execution logic, reducing turnaround time by **60%** while increasing release stability under high-volume usage scenarios.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **Python automation pipelines** processing 100K+ trajectory logs, supporting repeatable environment provisioning and performance diagnostics across iterative experimental runs.
- Produced structured troubleshooting artifacts and root-cause notes, improving incident response quality and cross-team remediation for unstable system behavior.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL cloud workflows** using OOP, AWS Glue, and S3, improving response time by **30%** and strengthening secure data movement across systems.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito across three OS environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **REST API and WebSocket services** supporting eight workflows and 2M+ synchronized records, improving service communication reliability and environment consistency for production operations.
- Automated deployments/testing with **Docker** and Jenkins, reducing deployment time by **50%** and improving rollback safety through repeatable, auditable release pipelines.

Skills

Cloud & Infra: **Azure** concepts (IAM, resource groups, policies), **AWS** fundamentals (EC2, S3, IAM, VPC), container familiarity (**Docker**, Kubernetes)
Automation & IaC: Python/PowerShell scripting, CI/CD concepts, module-based deployment mindset, version-controlled infrastructure workflows
Security & Governance: encryption/access-control basics, configuration drift awareness, tagging/governance practices, compliance-minded operations (FFIEC/NIST/SOC/CIS familiarity)
Data & APIs: PostgreSQL, MySQL, SQL Server concepts, REST APIs, ETL orchestration with AWS Glue/S3
Engineering: Git/GitHub, unit/integration testing, root-cause analysis, cross-functional collaboration

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Implemented **containerized cloud workflows** for 50K+ users and 1M+ transactions with scalable deployment patterns, secure integrations, and reliable service communication across distributed components.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Built **operational automation services** supporting 100+ daily appointments, adding monitoring-backed controls that improved retrieval latency by **2x** and strengthened production support readiness.

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Developed a **policy-aware data workflow engine** across 10+ databases with schema validation and structured execution checks to improve consistency and governance of backend operations.